

Conditionals and Control Flow in Java

else Statement

The `else` statement executes a block of code when the condition inside the `if` statement is `false`. The `else` statement is always the last condition.

```
boolean condition1 = false;

if (condition1){
    System.out.println("condition1 is true");
}
else{
    System.out.println("condition1 is not true");
}

// Prints: condition1 is not true
```

else if Statements

`else - if` statements can be chained together to check multiple conditions. Once a condition is `true`, a code block will be executed and the conditional statement will be exited.

There can be multiple `else - if` statements in a single conditional statement.

```
int testScore = 76;
char grade;

if (testScore >= 90) {
    grade = 'A';
} else if (testScore >= 80) {
    grade = 'B';
} else if (testScore >= 70) {
    grade = 'C';
} else if (testScore >= 60) {
    grade = 'D';
} else {
    grade = 'F';
}

System.out.println("Grade: " + grade); //
Prints: C
```

if Statement

An `if` statement executes a block of code when a specified boolean expression is evaluated as `true`.

```
if (true) {  
    System.out.println("This code  
executes");  
}  
// Prints: This code executes  
  
if (false) {  
    System.out.println("This code does  
not execute");  
}  
// There is no output for the above  
statement
```

Nested Conditional Statements

A nested conditional statement is a conditional statement nested inside of another conditional statement. The outer conditional statement is evaluated first; if the condition is `true`, then the nested conditional statement will be evaluated.

```
boolean studied = true;  
boolean wellRested = true;  
  
if (wellRested) {  
    System.out.println("Best of luck  
today!");  
    if (studied) {  
        System.out.println("You are prepared  
for your exam!");  
    } else {  
        System.out.println("Study before your  
exam!");  
    }  
}  
  
// Prints: Best of luck today!  
// Prints: You are prepared for your exam!
```

 **Print**